5.2 Entropy (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.2 Entropy (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

The feasibility of a reaction is determined by the relationship between enthalpy change, and entropy change.

Write the equation that shows the relationship between ΔG, ΔH and ΔS.

[1 mark]

Question 1b

What is the requirement for a reaction to be *feasible*?

[1 mark]

Question 1c

Chlorine can be formed in the following reversible reaction

$$4HCl(g) + O_2(g) \rightleftharpoons 2Cl_2(g) + 2H_2O(g)$$

Use the following data to calculate the standard entropy change, $\Delta S^{\ominus}$ for this reaction.

	HCl (g)	O_2 (g)	Cl_2 (g)	H_2O (g)
$S^{\ominus}$ / J K^{-1} mol^{-1}	187	205	223	189

The equation to calculate standard entropy change is:

$$\Delta S^{\ominus} = \Sigma S_{products} - \Sigma S_{reactants}$$

[3 marks]

Question 1d

The standard enthalpy change for the reaction in (c) is -116 kJ mol^{-1}

Calculate the minimum temperature at which this reaction is feasible.

[2 marks]

Question 2a

Ammonium chloride, NH_4Cl, can dissociate to form ammonia, NH_3, and hydrogen chloride, HCl.

$$NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$$

Explain whether entropy increases or decreases in this reaction.

[3 marks]

Question 2b

At 298K, ΔH = +176 kJ mol^{-1} and ΔS = +285 J mol^{-1} K^{-1}.

Calculate ΔG for this reaction at 298K.

[3 marks]

Question 2c

Explain whether the reaction is feasible under standard conditions.

[2 marks]

Question 2d

State and explain how the feasibility of the reaction will change with increasing temperature.

[2 marks]

Question 3a

The following equation represents the melting of ice.

$$H_2O\,(s) \rightarrow H_2O\,(l) \qquad \Delta H^{\ominus} = +6.03\ \text{kJ mol}^{-1} \quad \Delta S^{\ominus} = +22.1\ \text{J K}^{-1}\,\text{mol}^{-1}$$

State the meaning of the symbol $^{\ominus}$ in $\Delta H^{\ominus}$.

[1 mark]

Question 3b

Explain whether the entropy of the system increases or decreases.

[3 marks]

Question 3c

Calculate the temperature at which $\Delta G^{\ominus} = 0$ for this process.

Show your working.

[3 marks]

Model Answers: Easy

1a

a) The equation that shows the relationship between ΔG, ΔH and ΔS *is:*

- $\Delta G = \Delta H_{reaction} - T\Delta S_{system}$; [1 mark]

[Total: 1 mark]

- This is a standard equation that you will use at some point in your exams
- Other variations of the equation that would be accepted are:
  - $\Delta H_{reaction} = \Delta G + T\Delta S_{system}$
  - $\Delta S_{system} = (\Delta H_{reaction} - \Delta G) / T$
  - $T\Delta S_{system} = \Delta H_{reaction} - \Delta G$

1b

b) The requirement for a reaction to be feasible is:

- ΔG is less than or equal to zero; [1 mark]

[Total: 1 mark]

- This is the set condition / requirement that allows you to perform most ΔG calculations
- You would also score the mark for the following answers:
  - *ΔG is less than zero (ΔG < 0)*
  - *ΔG is equal to zero (ΔG = 0)*
  - *ΔG is negative*

1c

c) To calculate the standard entropy change:

- ([2 × 223] + [2 × 189]) - (205 + [4 × 187]); [1 mark]

- 824 – 953; [1 mark]

- −129 (J K^{-1} mol^{-1}); [1 mark]

[Total: 3 marks]

- You are not usually given the equation, this is one of the ones you need to learn
- Make sure you consider the number of moles of each substance there are and multiply this by the standard entropy change value
- Your answer must be negative, you will only score 1 mark for an answer of + 129 J K^{-1} mol^{-1}

1d

d) To calculate the temperature at which the reaction is feasible:

- $T = \frac{\Delta H}{\Delta S} = \frac{-116 \text{ x } 1000}{-129}$;[1 mark]
- 899 K; [1 mark]

[Total: 2 marks]

- From questions (a) and (b) we know:
  - For a reaction to be feasible ΔG should be less than 0
  - $\Delta G = \Delta H - T\Delta S$
- Therefore, $\Delta G = 0$, so $\Delta H - T\Delta S = 0$
  - This can be rearranged to: T= $\Delta H \div \Delta S$
- **Remember**: Don't forget to convert enthalpy change into J K^{-1} mol^{-1} by multiplying by 1000

2a

a) Entropy in the reaction:

- Increases; [1 mark]
- (Because) a gas is formed from a solid; [1 mark]
- Gases are more disordered / random / chaotic; [1 mark]

[Total: 3 marks]

- **Remember:** Entropy is a measure of disorder
  - The higher the entropy, the more disorder
- There is an increase in entropy from solid to liquid and liquid to gas, because the

molecules become more disordered as you move from one state to another

2b

b) To find ΔG:

- 285 ÷ 1000 = 0.285 kJ mol^{-1} K^{-1} ;[1 mark]
- ΔG = (+176) - (298 x 0.285); [1 mark]
- ΔG = 91.1 kJ mol^{-1}; [1 mark]

[Total: 3 marks]

- To find ΔG the following equation is used:
  - $\Delta G = \Delta H_{reaction} - T\Delta S_{system}$
  - First, you need to ensure the value for ΔS is converted from J mol^{-1} K^{-1} into kJ mol^{-1} K^{-1} by dividing by 1000
  - Then, it is a case of substituting the numbers into the Gibbs free energy equation

2c

c) The reaction is:

- Not feasible; [1 mark]
- (Because) $\Delta G^{\ominus}$ is positive; [1 mark]

[Total: 2 marks]

- **Remember**
  - When $\Delta G^{\ominus}$ is **negative**, the reaction is **feasible** and likely to occur
  - When $\Delta G^{\ominus}$ is **positive**, the reaction is **not feasible** and unlikely to occur

2d

d) With increasing temperature, the feasibility will:

- Increase / the reaction will become (more) feasible; [1 mark]
- (Because) ΔG will decrease; [1 mark]

[Total: 2 marks]

- When ΔH and ΔS are both positive, the reaction will become feasible after a certain temperature because $T\Delta S$ will be greater than ΔH
- Make sure you learn whether reactions are feasible based on the + or - of ΔG and ΔS

3a

a) The meaning of the symbol $^{\ominus}$ in $\Delta H^{\ominus}$ is:

- Standard conditions; [1 mark]

[Total: 1 mark]

- $^{\ominus}$ indicates that the value is true at standard conditions
- You can also score the mark by giving these conditions
  - Pressure = 101 kPa
  - Temperature = 298 K

3b

b) The entropy of the system:

- Increases; [1 mark]
- Because a liquid is formed; [1 mark]
- (Liquids are) more disordered / chaotic (than solids); [1 mark]

[Total: 3 marks]

- **Remember:** The more disordered a system is, the higher the entropy
- Gases have a higher entropy than liquids, and liquids have a higher entropy than solids due to the particles becoming more disordered from a solid to a gas

3c

c) The temperature at which $\Delta G^{\ominus}$ = 0 for this process is:

- T = $\frac{\Delta H}{\Delta S}$; [1 mark]
- T = $\frac{6.03 \times 1000}{22.1}$; [1 mark]
- = 273 K; [1 mark]

[Total: 3 marks]

- ΔS and ΔH values need to have the same units
- The value for ΔH can be converted from kJ mol^{-1} K^{-1} to J mol^{-1} K^{-1} by multiplying by 1000
- Alternatively, the value for ΔS can be converted from J mol^{-1} K^{-1} into kJ mol^{-1} K^{-1} by dividing by 1000
- In this example, ΔH has been converted

Calculate the minimum temperature at which the reaction becomes spontaneous. Show your working.

Temperature = ... K

How did you do?

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